CODORYTHM

250 REAL-WORLD ARCHITECTURE CASE STUDIES

Beginner to Intermediate Systems

Comprehensive System Design Learning Framework

Document Overview

This document presents Codorythm's comprehensive library of 250 real-world architecture case studies, organized into 13 major categories covering mainstream systems that form the foundation of modern internet infrastructure.

- 13 Major Categories: Web, Communication, Social, Media, E-Commerce, Finance, Mobility, Search, Developer Tools, Cloud, Data, AI/ML, Enterprise
- 250 Complete Systems: From URL shorteners to global SaaS platforms
- 8 Learning Modes: Understand, Rebuild, Break, Scale, Optimize, Interview, Implement, Architect
- 18-Component Template: Consistent, comprehensive case study framework
- Knowledge Graph Integration: Connect concepts to real-world applications
- Foundational Focus: Beginner and intermediate level difficulty

How to Use This Document

- Browse by category to find systems relevant to your interest
- Choose case studies matching your current skill level
- Study in the order listed within categories (building complexity)
- Apply the 8 learning modes progressively for mastery
- Connect to the System Design Knowledge Graph for concept understanding

CATEGORY A: WEB & INTERNET SYSTEMS (Cases 1-20)

Foundational internet infrastructure systems serving as building blocks for larger platforms

#	Case Study
1	Design a URL Shortener (Bitly)
2	Design a Pastebin
3	Design a Web Hosting Platform
4	Design a Domain Registration System
5	Design a DNS Service
6	Design a Reverse Proxy
7	Design a Load Balancer
8	Design a CDN
9	Design a File Upload Service
10	Design a File Download Service
11	Design a Static Website Hosting Platform
12	Design a Web Analytics Platform
13	Design a Website Monitoring System
14	Design an API Gateway
15	Design an API Rate Limiter
16	Design a CAPTCHA Service
17	Design a URL Preview Service
18	Design a Link Preview Generator

19	Design a Webhook Delivery Platform
20	Design an HTTP Proxy

Key Concepts Covered:

- HTTP/HTTPS protocol and DNS resolution
- Load balancing and request routing
- Caching strategies and TTL management
- API design and rate limiting
- File upload/download optimization

CATEGORY B: COMMUNICATION & COLLABORATION (Cases 21-40)

Real-time communication and collaborative systems connecting billions of users

#	Case Study
21	Design WhatsApp
22	Design Telegram
23	Design Signal
24	Design Slack
25	Design Discord
26	Design Microsoft Teams
27	Design Email System
28	Design Gmail
29	Design Push Notification System
30	Design SMS Gateway
31	Design In-App Notification System
32	Design Presence System
33	Design Online/Offline Status System
34	Design Group Chat
35	Design End-to-End Encrypted Messaging
36	Design Collaborative Document Editing
37	Design Google Docs-like Collaboration
38	Design Real-Time Whiteboard
39	Design Video Calling Platform
40	Design Zoom-like Meeting Platform

Key Concepts Covered:

- Real-time message delivery and ordering
- Presence and online status tracking
- End-to-end encryption and security
- WebSocket and persistent connections
- Message queuing and acknowledgment systems

CATEGORY C: SOCIAL MEDIA (Cases 41-60)

Large-scale social networks with billions of users, connections, and content

#	Case Study
41	Design Instagram
42	Design Facebook
43	Design Twitter/X
44	Design LinkedIn
45	Design Reddit
46	Design Snapchat
47	Design TikTok
48	Design Pinterest
49	Design Threads-like Social Platform
50	Design Social Media Feed
51	Design News Feed Ranking System
52	Design Like/Reaction System
53	Design Comment System
54	Design Follow/Followers System
55	Design Social Graph
56	Design Trending Topics System
57	Design Hashtag Search
58	Design Social Media Notification System
59	Design Story/Status System
60	Design Social Media Content Moderation Platform

Key Concepts Covered:

- Feed ranking and personalization algorithms
- Graph databases for social connections
- Real-time notification systems
- Content moderation and filtering
- Hashtag and trending topic indexing

CATEGORY D: VIDEO, AUDIO & MEDIA (Cases 61-80)

Media streaming and processing platforms delivering petabytes of content daily

#	Case Study
61	Design Netflix
62	Design YouTube
63	Design Spotify
64	Design Twitch
65	Design TikTok Video Platform
66	Design Instagram Reels
67	Design YouTube Shorts
68	Design Video Streaming Platform
69	Design Live Streaming Platform
70	Design Video Upload Platform
71	Design Video Transcoding Pipeline
72	Design Video Encoding Platform
73	Design Video CDN
74	Design Video Recommendation Engine
75	Design Movie Recommendation System

76	Design Music Recommendation System
77	Design Podcast Platform
78	Design Music Streaming Platform
79	Design Digital Asset Management System
80	Design Image Processing Pipeline

Key Concepts Covered:

- Video transcoding and adaptive bitrate streaming
- CDN distribution and edge caching
- Recommendation algorithms and personalization
- Live streaming infrastructure
- Media processing pipelines

CATEGORY E: E-COMMERCE (Cases 81-100)

Online retail platforms processing trillions in transactions annually

#	Case Study
81	Design Amazon
82	Design Flipkart
83	Design eBay
84	Design Shopify
85	Design E-Commerce Platform
86	Design Product Catalog
87	Design Product Search
88	Design Shopping Cart
89	Design Wishlist
90	Design Inventory Management System
91	Design Order Management System
92	Design Order Tracking System
93	Design Pricing Engine
94	Design Discount/Coupon System
95	Design Product Recommendation System
96	Design Seller Marketplace
97	Design Flash Sale System
98	Design Black Friday Sale System
99	Design Auction Platform
100	Design Review & Rating System

Key Concepts Covered:

- Inventory management and stock tracking
- Dynamic pricing and discount strategies
- Shopping cart and checkout systems
- Order fulfillment and tracking
- Product recommendation and search ranking

CATEGORY F: PAYMENTS & FINANCIAL SYSTEMS (Cases 101-120)

Critical financial infrastructure handling money flow and transactions safely

#	Case Study
101	Design Stripe
102	Design PayPal
103	Design Razorpay-like Payment Gateway
104	Design UPI-like Payment System
105	Design Digital Wallet
106	Design Banking System
107	Design Payment Processing System
108	Design Payment Ledger
109	Design Double-Entry Accounting System
110	Design Transaction Processing System
111	Design Payment Reconciliation System
112	Design Settlement System
113	Design Refund System
114	Design Recurring Payment System
115	Design Subscription Billing System
116	Design Fraud Detection System
117	Design Credit Card Processing System
118	Design Stock Trading Platform
119	Design Cryptocurrency Exchange
120	Design Financial Notification System

Key Concepts Covered:

- Payment processing and authorization
- Transaction atomicity and consistency
- Fraud detection and prevention
- Reconciliation and settlement
- Double-entry accounting and ledger systems

CATEGORY G: MOBILITY, MAPS & LOGISTICS (Cases 121-140)

Real-time location services and logistics platforms connecting services to customers

#	Case Study
121	Design Uber
122	Design Ola
123	Design Lyft
124	Design Google Maps
125	Design Waze
126	Design Ride Matching System
127	Design Driver Location Tracking
128	Design Real-Time Vehicle Tracking
129	Design Food Delivery Platform
130	Design Swiggy
131	Design Zomato
132	Design DoorDash
133	Design Delivery Partner Matching
134	Design Route Optimization System
135	Design Logistics Management Platform

136	Design Fleet Management System
137	Design Package Tracking System
138	Design Warehouse Management System
139	Design Geofencing Platform
140	Design Real-Time Geospatial Search

Key Concepts Covered:

- Geospatial indexing and proximity searches
- Real-time location tracking and updates
- Matching and assignment algorithms
- Route optimization
- Geofencing and location-based triggers

CATEGORY H: SEARCH, DISCOVERY & RECOMMENDATION (Cases 141-160)

Information retrieval and personalization systems helping users find what they need

#	Case Study
141	Design Google Search
142	Design Bing
143	Design Elasticsearch-like Search Engine
144	Design Web Crawler
145	Design Search Autocomplete
146	Design Spell Correction System
147	Design Image Search
148	Design Video Search
149	Design Product Search
150	Design Distributed Search Engine
151	Design Recommendation Engine
152	Design Netflix Recommendation Engine
153	Design Spotify Recommendation Engine
154	Design Amazon Recommendation Engine
155	Design Personalized Feed Ranking
156	Design Ad Recommendation System
157	Design Job Recommendation System
158	Design Content Discovery Platform
159	Design Similarity Search Engine
160	Design Semantic Search System

Key Concepts Covered:

- Web crawling and indexing
- Inverted indexes and search ranking
- Recommendation algorithms (collaborative, content-based)
- Personalization and A/B testing
- Search autocomplete and query understanding

CATEGORY I: DEVELOPER & ENGINEERING PLATFORMS (Cases 161-180)

Developer-focused infrastructure enabling software development and deployment

#	Case Study
161	Design GitHub
162	Design GitLab
163	Design Bitbucket
164	Design Git Hosting Platform
165	Design CI/CD Platform
166	Design Jenkins-like Build System
167	Design GitHub Actions-like Platform
168	Design Code Execution Platform
169	Design Online Coding Judge
170	Design Online IDE
171	Design Cloud IDE
172	Design Package Registry
173	Design Docker Container Registry
174	Design Artifact Repository
175	Design Feature Flag Platform
176	Design Configuration Management Platform
177	Design Secrets Management Platform
178	Design Developer Portal
179	Design Internal Developer Platform
180	Design Distributed Job Scheduler

Key Concepts Covered:

- Version control and Git operations at scale
- CI/CD pipelines and build automation
- Container registry and artifact management
- Secrets and configuration management
- Job scheduling and execution

CATEGORY J: CLOUD & INFRASTRUCTURE (Cases 181-200)

Cloud infrastructure and services providing compute, storage, and networking

#	Case Study
181	Design Amazon S3-like Object Storage
182	Design Google Drive
183	Design Dropbox
184	Design Google Cloud Storage-like Platform
185	Design Distributed File System
186	Design Distributed Cache
187	Design Redis-like Distributed Key-Value Store
188	Design Memcached-like Cache
189	Design Distributed Database
190	Design Apache Kafka-like Messaging Platform
191	Design RabbitMQ-like Message Broker
192	Design Cloud Load Balancer
193	Design Global Load Balancer
194	Design CDN Infrastructure
195	Design Cloud DNS

196	Design Serverless Platform
197	Design Function-as-a-Service Platform
198	Design Container Orchestration Platform
199	Design Kubernetes-like Platform
200	Design Cloud Monitoring Platform

Key Concepts Covered:

- Object storage and file systems
- Distributed caching and key-value stores
- Message brokers and event streaming
- Container orchestration and serverless
- Monitoring and observability infrastructure

CATEGORY K: DATA ENGINEERING & ANALYTICS (Cases 201-215)

Data platforms processing petabytes of data for business intelligence and analytics

#	Case Study
201	Design Data Warehouse
202	Design Data Lake
203	Design Data Lakehouse
204	Design ETL Platform
205	Design ELT Platform
206	Design Real-Time Data Pipeline
207	Design Apache Spark-like Processing Platform
208	Design Apache Flink-like Streaming Platform
209	Design Change Data Capture Platform
210	Design Data Catalog
211	Design Data Lineage Platform
212	Design Data Quality Platform
213	Design Event Analytics Platform
214	Design Real-Time Analytics Platform
215	Design Business Intelligence Platform

Key Concepts Covered:

- ETL/ELT pipelines and data transformation
- Data warehouse and lake architectures
- Stream processing and real-time analytics
- Data quality and validation
- Change data capture and CDC patterns

CATEGORY L: AI & MACHINE LEARNING SYSTEMS (Cases 216-235)

AI infrastructure for training, serving, and managing machine learning models

#	Case Study
216	Design ChatGPT-like AI Platform
217	Design LLM Inference Platform
218	Design AI Model Gateway
219	Design Multi-Model AI Router

220	Design GPU Inference Cluster
221	Design AI Model Serving Platform
222	Design RAG Platform
223	Design Enterprise RAG System
224	Design Vector Database
225	Design Semantic Search Platform
226	Design AI Agent Platform
227	Design Multi-Agent System
228	Design AI Memory System
229	Design AI Tool-Calling Platform
230	Design AI Observability Platform
231	Design AI Evaluation Platform
232	Design LLM Cost Optimization Platform
233	Design AI Recommendation Engine
234	Design Real-Time Fraud Detection ML Platform
235	Design ML Training Platform

Key Concepts Covered:

- Model training and distributed training
- Model serving and inference optimization
- Vector databases and embeddings
- RAG (Retrieval-Augmented Generation) systems
- Multi-agent orchestration and tool calling

CATEGORY M: ENTERPRISE & SaaS (Cases 236-250)

Enterprise software platforms serving organizations and businesses globally

#	Case Study
236	Design Multi-Tenant SaaS Platform
237	Design Enterprise CRM
238	Design Enterprise ERP
239	Design Learning Management System
240	Design Online Examination Platform
241	Design HR Management Platform
242	Design Payroll System
243	Design Recruitment Platform
244	Design Customer Support Platform
245	Design Enterprise Search Platform
246	Design Audit Management Platform
247	Design Document Management Platform
248	Design Enterprise Notification Platform
249	Design Multi-Region Enterprise SaaS
250	Design Global Enterprise Platform

Key Concepts Covered:

- Multi-tenancy and data isolation
- Enterprise security and compliance
- Workflow automation and orchestration
- Role-based access control (RBAC)
- Enterprise SaaS scalability patterns

8 LEARNING MODES FOR MASTERY

Each case study can be approached through 8 distinct learning modes, allowing deep engagement and progressive skill development.

1. UNDERSTAND

Deep dive into how the real system works and why it was designed that way.

- Study the business problem and context
- Learn functional requirements (features)
- Understand non-functional requirements (scale, latency, availability)
- Analyze the overall architecture
- Trace data flow for typical requests
- Understand technology choices and trade-offs

2. REBUILD

Build a similar system from scratch based on requirements.

- Select appropriate components and technologies
- Design API contracts and interfaces
- Design database schemas
- Design queue and caching strategies
- Create complete architecture diagram
- Justify all major decisions

3. BREAK

Inject failures and observe how the system responds.

- Simulate database failures and recovery
- Test cache failures and degradation
- Simulate service crashes
- Test network partitions and latency
- Simulate region failures
- Observe failure impact and mitigation

4. SCALE

Handle exponential growth by modifying the architecture.

- Increase traffic by 10x and identify bottlenecks
- Scale each component appropriately
- Increase traffic by 100x
- Re-architect for massive scale
- Increase traffic by 1,000x
- Design for planet-scale

5. OPTIMIZE

Improve the system across multiple dimensions simultaneously.

- Reduce infrastructure costs
- Reduce latency and improve response times
- Improve availability and reliability
- Increase throughput
- Optimize resource utilization
- Measure trade-offs between improvements

6. INTERVIEW

Practice with AI-driven real-time feedback.

- AI interviewer asks clarifying questions
- Address architecture challenges
- Defend design decisions
- Respond to failure scenario questions
- Justify trade-off choices
- Receive real-time scoring and feedback

7. IMPLEMENT

Write production-quality code for key components.

- Implement REST APIs
- Build message queue producers and consumers
- Design and code cache implementations
- Write database operations
- Build job schedulers and workers
- Create monitoring and alerting

8. ARCHITECT

Design the complete architecture with no predefined solution.

- Principal-level design challenge
- Create architecture from first principles
- Address all requirements and constraints
- Justify every architectural decision
- Demonstrate deep mastery of concepts
- Ready for senior-level interviews

STANDARD CASE STUDY TEMPLATE

Every case study follows an 18-component framework ensuring consistency and comprehensive coverage.

- 01 — Story: What problem did the company solve?
- 02 — Requirements: Functional + non-functional specifications
- 03 — Scale: Users, traffic, storage, latency, availability targets
- 04 — Naive Architecture: Simple starting design
- 05 — Problems: Where does the naive design break?
- 06 — Architecture Evolution: V1 → V2 → V3 → V4 → Global
- 07 — Components: Detailed explanation of each part
- 08 — Data Flow: Exactly what happens per request
- 09 — Failure Scenarios: Systematic failure analysis
- 10 — Scaling: Handle 10× / 100× / 1,000× traffic
- 11 — Trade-offs: Why architecture A instead of B?
- 12 — Cost: Infrastructure and operational costs
- 13 — Security: Threats and mitigation strategies
- 14 — Observability: Logs, metrics, traces, alerting
- 15 — Simulation: Interactive architecture playground
- 16 — Design Challenge: 'Now you design it'
- 17 — AI Interview: AI acts as real interviewer
- 18 — Architecture Score: Multi-dimensional scoring

Scoring Dimensions

- Requirements fulfillment and completeness
- Scalability and performance characteristics
- Reliability and fault tolerance
- Latency and response time optimization
- Data design and consistency model
- Security posture and threat mitigation
- Cost efficiency and resource optimization
- Trade-off analysis and justification
- Overall architecture quality

NETFLIX: COMPLETE CASE STUDY DEEP DIVE

Netflix demonstrates how a single mainstream case study can be decomposed into comprehensive learning journey across 10 parts.

Part 1: Understanding Netflix

- Business model: Subscription-based video streaming
- Market: 250M+ subscribers across 190+ countries
- Content: Movies, series, documentaries, stand-up
- Challenges: License management, content delivery, personalization

Part 2: Core Architecture

- Client layer: Smart TV, mobile, web browsers
- API gateway: Route requests, rate limit, authenticate
- Microservices: User service, content service, playback service
- Databases: User data, content metadata, viewing history
- Cache layer: Redis for frequently accessed data
- Event streaming: Track user activity in real-time

Part 3: Video Processing Pipeline

Content Upload → Encoding → Transcoding → Multi-bitrate → Storage → CDN → User Device

- Content acquisition and licensing management
- Video upload infrastructure
- Encoding at multiple resolutions
- Adaptive bitrate transcoding (1080p, 720p, 480p, etc.)
- Multi-format storage optimization
- CDN distribution at scale

Part 4: Content Delivery System

- Why distribute content geographically?
- Control-plane: Decides what content goes where
- Data-plane: Serves content to end users
- CDN edge servers in strategic locations
- Cache invalidation and refresh strategies
- Request routing and load balancing

Part 5: Recommendation System

User Activity → Pipeline → Processing → ML Models → Personalized Recommendations

- Real-time event collection and streaming
- Batch data processing and aggregation
- Feature engineering from user behavior
- ML model training and serving

- Ranking and personalization strategies

Part 6: Scaling to 10x Users

Question: What happens when Netflix gets 10x more users?

- Database sharding and replication strategy
- Cache partitioning and distribution
- Service replication and auto-scaling
- CDN capacity expansion
- ML pipeline infrastructure growth

Part 7: Failure Injection

- Database failure → Fallback to read replicas
- Cache failure → Graceful degradation to database
- Service failure → Load balanced failover
- Region failure → Multi-region automatic failover
- CDN failure → Origin server fallback

Part 8: Chaos Engineering

- Deliberately inject latency spikes
- Simulate cache stampede scenarios
- Test database connection pool exhaustion
- Test network partition handling
- Measure system resilience and recovery time

Part 9: Architecture Decisions

Evaluate alternatives and justify choices:

- Centralized CDN vs. Third-party CDN vs. Own infrastructure?
- Monolithic vs. Microservices architecture?
- Cache-aside vs. Write-through caching?
- Sync recommendation generation vs. Async batch processing?

Part 10: Student Design Challenge

- Design Netflix architecture from scratch
- No predefined solution provided
- Must address all non-functional requirements
- Document all design decisions
- Prepare for AI interviewer evaluation

KNOWLEDGE GRAPH CONNECTIONS

Connect case studies back to the System Design Knowledge Graph, enabling students to see how concepts manifest in real systems.

Example 1: Apache Kafka Concept

Where is Kafka used in these 250 case studies?

- Netflix: Content activity tracking for recommendations
- Uber: Real-time event streaming for ride updates
- LinkedIn: User activity and notification pipeline
- Slack: Message delivery and presence updates
- DoorDash: Order status updates and delivery tracking

- Stripe: Payment event processing
- Social media platforms: Feed updates and notifications

Example 2: Distributed Cache Concept

Where are distributed caches used in these 250 case studies?

- Netflix: Content metadata and viewing preferences
- Amazon: Product catalog and shopping cart data
- Facebook/Instagram: Social graph and friend lists
- Uber: Driver location and available rides
- Google Search: Query results and page rankings
- E-commerce: Inventory and pricing data
- Payment systems: Transaction verification data

Example 3: Geospatial Indexing Concept

Where is geospatial indexing used in these 250 case studies?

- Uber: Find nearby drivers and passengers
- Google Maps: Route finding and traffic updates
- Zomato/Swiggy: Locate restaurants and delivery partners
- DoorDash: Find delivery partners and restaurants
- Logistics: Track vehicle and package locations
- Social apps: Location-based friend discovery

RECOMMENDED LEARNING PROGRESSION

Suggested study order for mastering all 250 case studies

Phase 1: Foundation (Cases 1-50)

Focus: Master fundamental internet and web architecture concepts

- Complete Category A (Web & Internet) thoroughly
- Study Category B (Communication) first 10 cases
- Choose 3-4 complex systems to deep dive
- Use UNDERSTAND and REBUILD modes
- Time: 4-6 weeks

Phase 2: Scale & Complexity (Cases 51-150)

Focus: Handle complexity and learn scaling strategies

- Complete Categories B-F (Communication, Social, Media, E-Commerce, Finance)
- Start using BREAK and SCALE modes
- Practice 5-6 design challenges
- Implement code for key components
- Time: 8-12 weeks

Phase 3: Specialization (Cases 151-250)

Focus: Deep expertise in specialized domains

- Complete Categories G-M (Mobility, Search, Developer Tools, Cloud, Data, AI, Enterprise)
- Choose domain specialization (e.g., payments, AI, data)
- Master OPTIMIZE and ARCHITECT modes
- Practice with AI interviewer extensively
- Time: 12-16 weeks

Total Time Investment: 24-34 weeks (6-8 months) for comprehensive mastery

CONCLUSION

The 250 case studies library provides a comprehensive, practical foundation for system design mastery. By studying real-world systems across 13 major categories, engaging through 8 distinct learning modes, and progressively applying the 18-component case study framework, you develop both breadth and depth in architectural thinking.

This foundation prepares you for:

- System Design interviews at top companies
- Real-world architectural decisions
- Scaling systems from thousands to billions of users
- Building reliable, resilient infrastructure
- Advancing to staff and principal architect roles

Remember: These 250 case studies are not endpoints, but waypoints on the journey to architectural mastery. The next 250 advanced and planet-scale systems await as you grow in expertise.